Limits in Finance

From discrete models to continuous risk optimization — the complete mathematics behind the project

THE QUESTION. A saver can keep money in a risk-free account paying $r = 3\%$ per year, or invest in a stock index with expected return $\mu = 10\%$ and volatility $\sigma = 20\%$. **What fraction f of the savings should go into the stock** so that wealth grows as fast as possible, while respecting the rule *"with 95% confidence, do not lose more than 15% in one year"*?

THE ANSWER (derived in Part V): $f_{opt} = 0.705$ — put **70.5% in the stock and 29.5% risk-free**.

THE METHOD. Real finance is *discrete* (interest paid monthly, prices quoted tick by tick, portfolios rebalanced daily). We let the step size go to zero — a **limit** — and obtain continuous models with closed-form expressions. Then **continuity** (the Intermediate Value Theorem) guarantees the optimal allocation exists, and bisection computes it.

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PARAMETERS USED THROUGHOUT.

Module	Parameters
1 Compounding	$P = 10{,}000$, $r = 10\%$, $t = 10$ years
2 Options	$S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$
3–4 Market	$\mu = 10\%$, $\sigma = 20\%$, $r = 3\%$, $T = 1$ year, $\alpha = 95\%$, loss limit $L = 15\%$, leverage cap $f_{max} = 2$

Part I — Mathematical toolkit

This part collects every theorem the project relies on, so that later parts can quote them.

1.1 Limits of sequences and continuity

Definition 1.1 (limit). A sequence (a_n) converges to a, written $a_n \to a$, if for every $\varepsilon > 0$ there is N such that $|a_n - a| < \varepsilon$ for all $n \geq N$.

Definition 1.2 (continuity). h is continuous at x if for every $\varepsilon > 0$ there is $\delta > 0$ with $|y - x| < \delta \Rightarrow |h(y) - h(x)| < \varepsilon$.

Lemma 1.3 (sequential continuity). h is continuous at x if and only if $x_n \to x$ implies $h(x_n) \to h(x)$. In words: **a limit can be moved inside a continuous function**,

$$\lim_{n\to\infty} h(x_n) = h\Big(\lim_{n\to\infty} x_n\Big).$$

We use Lemma 1.3 constantly, typically with $h = \exp$: to find $\lim e^{x_n}$ it suffices to find $\lim x_n$.

Theorem 1.4 (squeeze; monotone convergence). (a) If $b_n \leq a_n \leq c_n$ and $b_n, c_n \to a$, then $a_n \to a$. (b) An increasing sequence that is bounded above converges (completeness of $\mathbb{R}$).

1.2 The exponential and the logarithm

Lemma 1.5 (logarithm bounds). For every $u > -1$, $u \neq 0$:

$$\frac{u}{1+u} \;<\; \ln(1+u) \;<\; u. \tag{1}$$

Proof. Let $\psi(u) = u - \ln(1+u)$. Then $\psi(0) = 0$ and $\psi'(u) = 1 - \frac{1}{1+u} = \frac{u}{1+u}$, which is negative on $(-1, 0)$ and positive on $(0, \infty)$; so ψ has a strict minimum at 0 and $\psi(u) > 0$ for $u \neq 0$. That is the right inequality. For the left one apply the right inequality to $v = -u/(1+u) > -1$: $\ln(1+v) = -\ln(1+u) < v = -u/(1+u)$. ∎

Lemma 1.6 (Taylor expansion). For $|u| < 1$,

$$\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots = u - \frac{u^2}{2} + O(u^3), \qquad e^u = 1 + u + \frac{u^2}{2} + O(u^3).$$

Notation: $O(u^k)$ means "bounded by a constant times $|u|^k$ for small u".

1.3 The derivative is a limit

Definition 1.7.

$$h'(x) = \lim_{\Delta \to 0} \frac{h(x+\Delta) - h(x)}{\Delta}.$$

The fraction is the *difference quotient* (slope of a secant line); its limit is the slope of the tangent.

Second-derivative test / concavity. If $h''(x) < 0$ for all x, then h is strictly concave, h' is strictly decreasing, and any point with $h'(x^*) = 0$ is the unique global maximum.

1.4 The Intermediate Value Theorem and the Extreme Value Theorem

Theorem 1.8 (Intermediate Value Theorem, IVT). Let $h : [a, b] \to \mathbb{R}$ be continuous with $h(a) < 0 < h(b)$ (or the reverse). Then there exists $c \in (a, b)$ with $h(c) = 0$.

Proof. Let $c = \sup\{x \in [a, b] : h(x) \le 0\}$, which exists by completeness. If $h(c) < 0$, continuity gives a neighbourhood to the right of c where $h < 0$, contradicting the definition of sup. If $h(c) > 0$, continuity gives a neighbourhood of c where $h > 0$, so a smaller number would be an upper bound, again a contradiction. Hence $h(c) = 0$. ∎

Geometric meaning: a continuous graph cannot get from below the axis to above it without crossing it. Continuity is essential — a function with a jump can skip over 0.

Theorem 1.9 (Extreme Value Theorem). A continuous function on a closed bounded interval $[a, b]$ attains a maximum and a minimum.

1.5 Bisection — the IVT as an algorithm

Algorithm. Start with $[a_0, b_0] = [a, b]$, $h(a)h(b) < 0$. For $k = 0, 1, 2, \ldots$: let $m_k = (a_k + b_k)/2$. If $h(a_k)h(m_k) < 0$ set $[a_{k+1}, b_{k+1}] = [a_k, m_k]$, otherwise $[a_{k+1}, b_{k+1}] = [m_k, b_k]$.

Theorem 1.10 (convergence of bisection). If h is continuous on $[a, b]$ with $h(a)h(b) < 0$, the midpoints m_k converge to a root c of h, and

$$|m_k - c| \le \frac{b-a}{2^{k+1}}.$$

Hence to reach accuracy tol it suffices to take $k \ge \log_2\big((b-a)/\mathrm{tol}\big)$ steps.

Proof. Each step keeps an interval on which h changes sign, so by the IVT every $[a_k, b_k]$ contains a root. (a_k) is increasing and bounded by b; (b_k) is decreasing and bounded by a; both converge (Thm 1.4b), and

since $b_k - a_k = (b-a)/2^k \to 0$ they have the same limit c. From $h(a_k)h(b_k) \le 0$ and continuity (Lemma 1.3), $h(c)^2 = \lim h(a_k)h(b_k) \le 0$, so $h(c) = 0$. Finally m_k and c both lie in $[a_k, b_k]$, and m_k is its midpoint. ∎

In our project, $[0, 2]$ with tolerance 10^{-12} needs $\log_2(2/10^{-12}) = 40.86$ — i.e. **41 steps**, exactly what the program prints.

1.6 Probability background

Normal distribution. $Z \sim N(0,1)$ has density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and CDF $\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \varphi$. $X \sim N(m, s^2)$ means $X = m + sZ$. Φ is continuous and strictly increasing, and $\Phi(-z) = 1 - \Phi(z)$.

Quantile. For $q \in (0,1)$ the quantile $z_q = \Phi^{-1}(q)$ is the unique solution of $\Phi(z) = q$. Existence follows from the IVT, uniqueness from strict monotonicity. We need $z_{0.95} = 1.644854$.

Quantiles and monotone maps. If ψ is strictly increasing then the q-quantile of $\psi(X)$ is ψ(q-quantile of X).

Lemma 1.11 (lognormal moments). If $X \sim N(m, s^2)$ then $E[e^X] = e^{m + s^2/2}$.

Proof. $E[e^X] = \int \frac{1}{s\sqrt{2\pi}} e^{x - \frac{(x-m)^2}{2s^2}}\,dx$. Completing the square, $x - \frac{(x-m)^2}{2s^2} = m + \frac{s^2}{2} - \frac{(x-m-s^2)^2}{2s^2}$, and the remaining integral is that of the $N(m+s^2, s^2)$ density, which equals 1. ∎

Convergence in distribution. $X_n \Rightarrow X$ if $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X.

Central Limit Theorem (de Moivre–Laplace / Lindeberg form). Sums of many small independent random steps, centred and scaled, converge in distribution to a normal law. In particular if $J_n \sim \mathrm{Bin}(n, p_n)$ with $np_n(1-p_n) \to \infty$, then $\frac{J_n - np_n}{\sqrt{np_n(1-p_n)}} \Rightarrow N(0,1)$.

Kolmogorov–Smirnov distance. $D(F,G) = \sup_x |F(x) - G(x)|$. If the limit F is continuous, $F_n \to F$ pointwise implies $D(F_n, F) \to 0$ (Pólya's theorem).

Berry–Esseen theorem. For a normalised sum of n i.i.d. steps with finite third moment, $D(F_n, \Phi) \le C/\sqrt{n}$.

Law of large numbers. Averages of i.i.d. variables converge to the mean: $\frac{1}{n}\sum_{k=1}^{n} Y_k \to E[Y]$.

Part II — Module 1: Continuous compounding

2.1 Model

Invest P at nominal annual rate r, compounded n times per year. Each period the balance is multiplied by $1 + r/n$; after t years there are nt periods, so

$$A_n = P\left(1 + \frac{r}{n}\right)^{nt}. \tag{2}$$

Question: what happens as compounding becomes more and more frequent, $n \to \infty$?

2.2 The limit

> **Theorem 2.1.** $\lim_{n\to\infty} P\left(1 + \frac{r}{n}\right)^{nt} = Pe^{rt}$ for every $r > 0, t > 0$.

Proof. Write $A_n = P\exp(x_n)$ with $x_n = nt\ln(1 + r/n)$. Apply Lemma 1.5 with $u = r/n$:

$$nt \cdot \frac{r/n}{1 + r/n} \;<\; x_n \;<\; nt \cdot \frac{r}{n} \quad\Longleftrightarrow\quad \frac{rt}{1 + r/n} \;<\; x_n \;<\; rt.$$

The left side tends to rt, so by the squeeze theorem $x_n \to rt$. Since $\exp$ is continuous, Lemma 1.3 gives $A_n = Pe^{x_n} \to Pe^{rt}$. ∎

> **Proposition 2.2 (more frequent compounding always pays more, but boundedly).** A_n is strictly increasing in n and $A_n < Pe^{rt}$ for every n.

Proof. Consider $\phi(x) = x\ln(1 + r/x)$ for real $x > 0$. With $u = r/x$, $\phi'(x) = \ln(1 + u) - \frac{u}{1+u} > 0$ by the left inequality in Lemma 1.5. So ϕ is increasing and $A_n = Pe^{t\phi(n)}$ increases with n. The upper bound is the right inequality $x_n < rt$ from the previous proof. ∎

So (A_n) is increasing and bounded: it converges by monotone convergence, and Theorem 2.1 identifies the limit.

2.3 How fast? The rate of convergence

Using Lemma 1.6 with $u = r/n$:

$$x_n = nt\left(\frac{r}{n} - \frac{r^2}{2n^2} + O(n^{-3})\right) = rt - \frac{r^2 t}{2n} + O(n^{-2}).$$

Let $\varepsilon_n = rt - x_n = \frac{r^2 t}{2n} + O(n^{-2})$. Then

$$Pe^{rt} - A_n = Pe^{rt}\left(1 - e^{-\varepsilon_n}\right) = Pe^{rt}\left(\varepsilon_n + O(\varepsilon_n^2)\right) = \boxed{Pe^{rt}\,\frac{r^2 t}{2n} + O\!\left(\frac{1}{n^2}\right)}. \tag{3}$$

The gap shrinks like $1/n$: doubling the compounding frequency roughly halves the remaining gain.

2.4 Numbers ($P = 10{,}000$, $r = 10\%$, $t = 10$: $Pe^{rt} = 10{,}000\,e = 27{,}182.82$)

compounding	n	A_n	gap $Pe^{rt} - A_n$	estimate (3)
yearly	1	25,937.42	1,245.39	1,359.14
quarterly	4	26,850.64	332.18	339.79
monthly	12	27,070.41	112.40	113.26
weekly	52	27,156.73	26.09	26.14
daily	365	27,179.10	3.72	3.72
hourly	8,760	27,182.66	0.155	0.155
every minute	525,600	27,182.82	0.0026	0.0026

The estimate is poor for $n = 1$ (where $r/n = 0.1$ is not small, so the $O(n^{-2})$ term matters) and becomes essentially exact as n grows — itself an illustration of an asymptotic limit. On a log–log plot the gap is a straight line of slope -1.

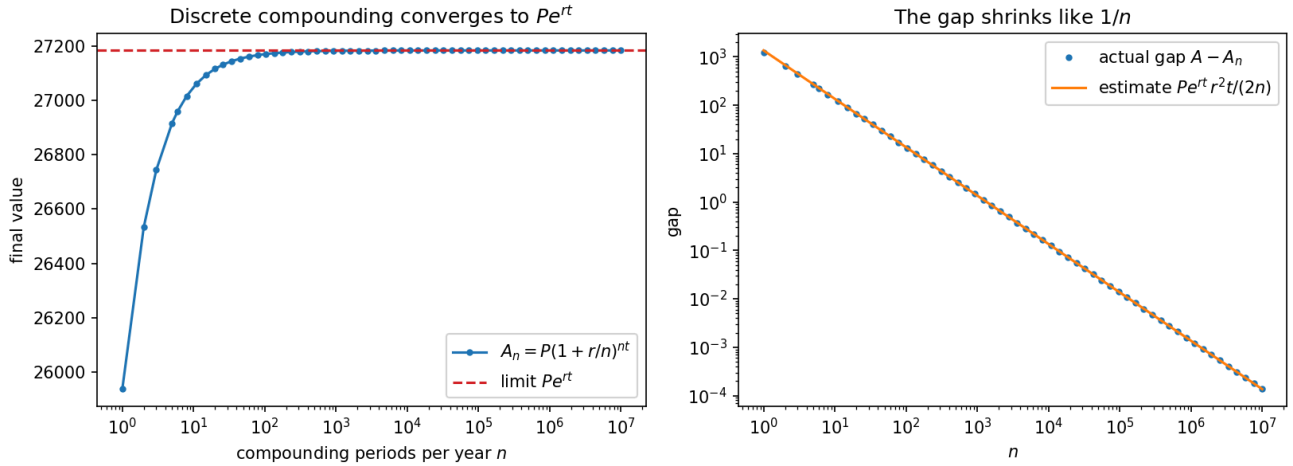


Figure 1. Left: A_n increases to the limit Pe^{rt}. Right: log–log plot of the gap, with the estimate $Pe^{rt}r^2t/(2n)$.

> **Real-life conclusion.** Switching from monthly to daily compounding gains 108.68; from daily to continuous only 3.72. Continuous compounding is the natural *idealisation* of frequent compounding, and its effective annual rate is $e^r - 1 = 10.517\%$ for $r = 10\%$. Modules 2–4 use e^{rt} as the growth of the risk-free asset.

Part III — Module 2: Binomial model → Black–Scholes

3.1 What is being priced

A **European call option** with strike K and maturity T gives the right (not the obligation) to buy the stock at price K at time T. Its value at maturity is the **payoff** $(S_T - K)^+ = \max(S_T - K, 0)$. A put pays $(K - S_T)^+$. Question: what is the fair price today?

3.2 One step: no-arbitrage pricing

In one step of length Δt, the stock moves from S to Su or Sd ($d < u$); money in the bank grows by $e^{r\Delta t}$. Let the option be worth V_u or V_d after the step. Build a portfolio of Δ shares and B in the bank that pays exactly the same in both states:

$$\Delta S u + B e^{r\Delta t} = V_u, \qquad \Delta S d + B e^{r\Delta t} = V_d.$$

Subtracting, $\Delta = \dfrac{V_u - V_d}{S(u-d)}$, and then $B = e^{-r\Delta t}\dfrac{uV_d - dV_u}{u-d}$. If the option were priced differently from the portfolio, one could buy the cheaper and sell the dearer for a riskless profit (an *arbitrage*). So

$$V = \Delta S + B = e^{-r\Delta t}\big[pV_u + (1-p)V_d\big], \qquad p = \frac{e^{r\Delta t} - d}{u - d}. \tag{4}$$

No arbitrage between stock and bank requires $d < e^{r\Delta t} < u$, i.e. $0 < p < 1$: p behaves like a probability, called the **risk-neutral probability**. Remarkably, the real-world drift μ does not appear.

3.3 N steps: the Cox–Ross–Rubinstein tree

Split $[0, T]$ into N steps, $\Delta t = T/N$, with $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$. After N steps with j up-moves the price is $Su^j d^{N-j} = Se^{\sigma\sqrt{\Delta t}(2j-N)}$. Applying (4) backwards through the tree (this is what the code does) is equivalent to the closed form

$$C_N = e^{-rT}\sum_{j=0}^{N}\binom{N}{j}p^j(1-p)^{N-j}\big(Su^j d^{N-j} - K\big)^+ = e^{-rT}\,E\big[(Se^{X_N} - K)^+\big],$$

where $X_N = \sigma\sqrt{\Delta t}\,(2J - N)$ and $J \sim \mathrm{Bin}(N, p)$ counts up-moves.

3.4 The limit $N \to \infty$

Step 1: expand p. With $\delta = \sqrt{\Delta t}$: $u = 1 + \sigma\delta + \tfrac{1}{2}\sigma^2\delta^2 + O(\delta^3)$, $d = 1 - \sigma\delta + \tfrac{1}{2}\sigma^2\delta^2 + O(\delta^3)$, $e^{r\Delta t} = 1 + r\delta^2 + O(\delta^4)$. Hence $u - d = 2\sigma\delta + O(\delta^3)$ and $e^{r\Delta t} - d = \sigma\delta + (r - \tfrac{1}{2}\sigma^2)\delta^2 + O(\delta^3)$, so

$$p = \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{\Delta t} + O(\Delta t).$$

(For $N = 100$: $p = 0.50750$, and the formula gives $0.5 + 0.03/0.4 \cdot 0.1 = 0.5075$.)

Step 2: mean and variance of X_N. Since $E[J] = Np$, $\mathrm{Var}(J) = Np(1-p)$:

$$E[X_N] = \sigma\sqrt{\Delta t}\,N(2p-1) = \sigma\sqrt{\Delta t}\,N\frac{r - \sigma^2/2}{\sigma}\sqrt{\Delta t} + O(\sqrt{\Delta t}) = (r - \tfrac{1}{2}\sigma^2)T + O(\sqrt{\Delta t}),$$

$$\mathrm{Var}(X_N) = 4\sigma^2\Delta t\,Np(1-p) = \sigma^2 T \cdot 4p(1-p) \;\to\; \sigma^2 T \quad (\text{since } p \to \tfrac{1}{2}).$$

Step 3: CLT. X_N is a sum of N independent small steps $\pm\sigma\sqrt{\Delta t}$. By the CLT,

$$X_N \Rightarrow X \sim N\big((r-\tfrac12\sigma^2)T,\ \sigma^2 T\big).$$

Step 4: pass to the limit in the price. The payoff $x \mapsto (Se^x - K)^+$ is continuous, and its growth is controlled ($E[e^{2X_N}]$ stays bounded), which is enough to conclude $E[(Se^{X_N}-K)^+] \to E[(Se^X-K)^+]$. Therefore $\lim_{N\to\infty} C_N = e^{-rT}E[(Se^X-K)^+]$.

3.5 Computing the limit: the Black–Scholes formula

Let $m = (r-\tfrac12\sigma^2)T$, $s = \sigma\sqrt{T}$, $k = \ln(K/S)$. The option pays only when $X > k$, so

$$E[(Se^X-K)^+] = S\,E\big[e^X \mathbf{1}_{X>k}\big] - K\,P(X>k).$$

Second term: $P(X>k) = P\big(Z > \tfrac{k-m}{s}\big) = \Phi\big(\tfrac{m-k}{s}\big) = \Phi(d_2)$, with $d_2 = \dfrac{\ln(S/K) + (r-\tfrac12\sigma^2)T}{\sigma\sqrt{T}}$.

First term: completing the square as in Lemma 1.11, e^x times the $N(m,s^2)$ density equals $e^{m+s^2/2}$ times the $N(m+s^2,s^2)$ density. Hence $E[e^X\mathbf{1}_{X>k}] = e^{m+s^2/2}\,\Phi\big(\tfrac{m+s^2-k}{s}\big) = e^{rT}\Phi(d_1)$, with $d_1 = d_2 + \sigma\sqrt{T}$.

Multiplying by e^{-rT}:

> $$C_{BS} = S\,\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r \pm \tfrac12\sigma^2)T}{\sigma\sqrt{T}}.$$
>
> For the put, use **put–call parity**: since $(x-K)^+ - (K-x)^+ = x - K$, we get $C - P = S - Ke^{-rT}$, so $P_{BS} = Ke^{-rT}\Phi(-d_2) - S\Phi(-d_1)$.

3.6 Numbers and rate of convergence

With $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$: $d_1 = 0.35$, $d_2 = 0.15$, $C_{BS} = 10.450584$.

N	1	2	5	10	50	100	500	1000
C_N	12.1623	9.5405	10.8059	10.2534	10.4107	10.4306	10.4466	10.4486
$\lvert C_N - C_{BS}\rvert$	1.7117	0.9101	0.3554	0.1972	0.0399	0.0200	0.0040	0.0020

The error behaves like c/N: the program gives $N\cdot|C_N - C_{BS}| \approx 2.00$ for $N = 100, 1000$ but ≈ 1.75 for $N = 101, 1001$. The constant depends on the parity of N because the position of the strike K relative to the tree's final nodes changes with N. This is the **even/odd oscillation** visible in Figure 2.

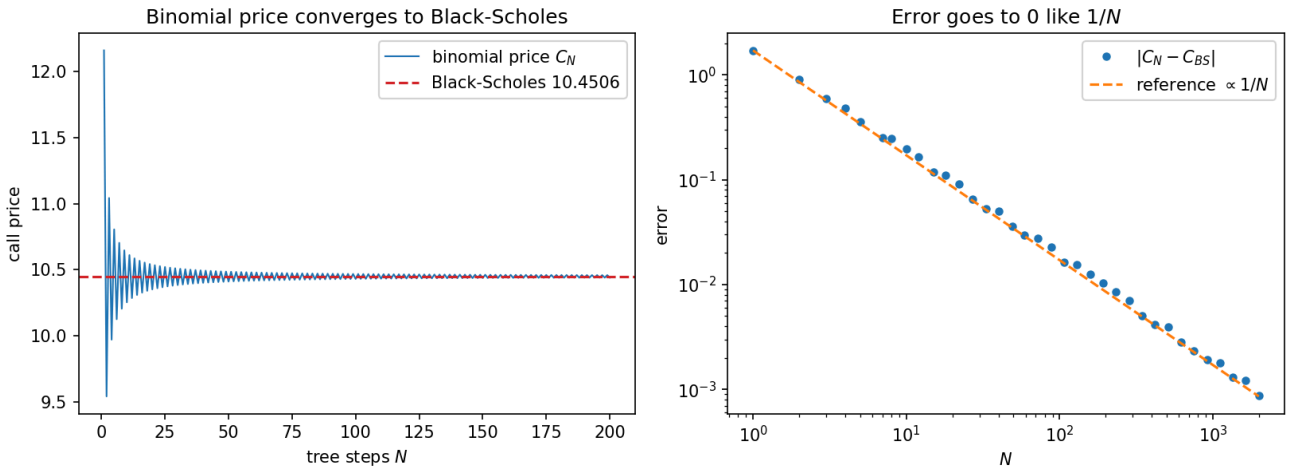


Figure 2. Left: binomial price vs. number of steps, oscillating towards Black–Scholes. Right: the error decays like $1/N$.

Part IV — Module 3: Random walk → Geometric Brownian Motion

4.1 The continuous model: Geometric Brownian Motion

The standard model of a stock price is

$$S_t = S_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma B_t \Big),$$

where B_t is Brownian motion ($B_t \sim N(0,t)$, independent increments). So the log-return is normal:

$$\ln \frac{S_T}{S_0} \sim N\Big((\mu - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T \Big). \tag{5}$$

Why $-\frac{1}{2}\sigma^2$? By Lemma 1.11, $E[S_T] = S_0 e^{(\mu-\sigma^2/2)T + \sigma^2 T/2} = S_0 e^{\mu T}$. So μ is the growth rate of the *average* price, while the typical (median) price grows only at $\mu - \frac{1}{2}\sigma^2$. This "volatility drag" returns in Module 4 as the $-\frac{1}{2}f^2\sigma^2$ term.

4.2 The discrete model: a biased coin-flip walk

Take n steps of length $\Delta t = T/n$. At each step the log-price moves up or down by $\sigma\sqrt{\Delta t}$, up with probability

$$p = \frac{1}{2} + \frac{1}{2}\cdot\frac{\mu - \sigma^2/2}{\sigma}\sqrt{\Delta t}.$$

(We need $0 \le p \le 1$, i.e. $\Delta t \le \sigma^2/(\mu-\sigma^2/2)^2 = 6.25$ years — always true here.) After n steps, with $J \sim \mathrm{Bin}(n,p)$ up-moves, $\ln(S_T/S_0) = \sigma\sqrt{\Delta t}\,(2J - n)$.

> **Proposition 4.1.** For every n:
>
> $$E\Big[\ln\frac{S_T}{S_0}\Big] = (\mu - \tfrac{1}{2}\sigma^2)T \ \text{(exactly)}, \qquad \mathrm{Var}\Big[\ln\frac{S_T}{S_0}\Big] = \sigma^2 T\Big(1 - \frac{(\mu-\sigma^2/2)^2}{\sigma^2}\Delta t\Big) \xrightarrow[\Delta t\to 0]{} \sigma^2 T.$$

Proof. Mean: $\sigma\sqrt{\Delta t}\,n(2p-1) = \sigma\sqrt{\Delta t}\,n\,\frac{\mu-\sigma^2/2}{\sigma}\sqrt{\Delta t} = (\mu - \frac{1}{2}\sigma^2)n\Delta t$. Variance: $4\sigma^2\Delta t\,np(1-p) = \sigma^2 T\cdot 4p(1-p)$ and $4p(1-p) = 1-(2p-1)^2$. ∎

Example $n = 12$: variance 0.039467 vs. limit 0.04. By the CLT the whole distribution converges: $\ln(S_T/S_0) \Rightarrow$ the GBM law (5). **The random walk becomes GBM in the limit $\Delta t \to 0$.**

4.3 Measuring the convergence exactly

We measure the distance between the walk's CDF F_n and the GBM CDF F by the Kolmogorov–Smirnov distance $D_n = \sup_x |F_n(x) - F(x)|$. Because F_n is a step function with jumps at the lattice points $x_k = \sigma\sqrt{\Delta t}(2k-n)$, and F is increasing, the supremum is attained at a jump, either just before it or at it:

$$D_n = \max_{0\le k\le n} \max\big(|F_n(x_k) - F(x_k)|,\ |F_n(x_k^-) - F(x_k)| \big).$$

This is computed from the exact binomial probabilities, so there is **no simulation noise**. Binomial coefficients like $\binom{25200}{12600} \approx 10^{7583}$ overflow a computer, so they are built in logarithms using $\ln\binom{n}{k} = \ln\binom{n}{k-1} + \ln\frac{n-k+1}{k}$.

4.4 Why exactly $1/\sqrt{n}$?

Upper bound: Berry–Esseen gives $D_n \le C/\sqrt{n}$. **Lower bound:** at a lattice point F_n jumps by $P(J=k)$, but the continuous F barely moves there, so F is at least half a jump away from F_n on one side: $D_n \gtrsim \frac{1}{2}\max_k P(J=k)$. By the local CLT (Stirling's formula), $\max_k P(J=k) \approx 1/\sqrt{2\pi np(1-p)}$. Therefore

$$D_n \approx \frac{1}{2\sqrt{2\pi np(1-p)}} \approx \frac{0.399}{\sqrt{n}}.$$

n (steps per year)	1	12	252	25,200
exact D_n	0.4257	0.1192	0.02530	0.002515
$1/(2\sqrt{2\pi np(1-p)})$	0.4353	0.1159	0.02514	0.002513

The prediction matches the computation almost perfectly, and explains the slope $-\frac{1}{2}$ on the log–log plot in Figure 3.

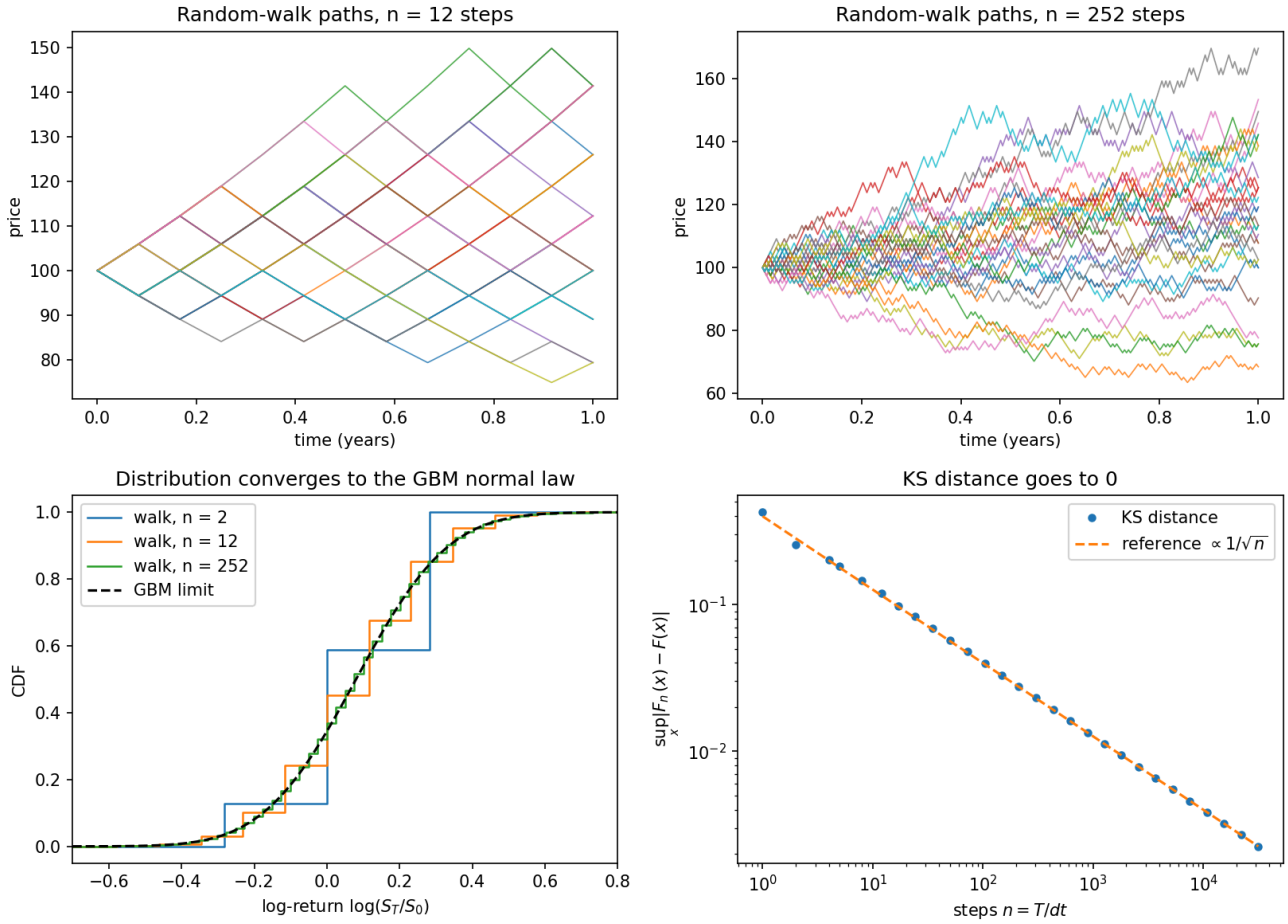


Figure 3. Top: walk paths with 12 and 252 steps. Bottom left: walk CDFs (steps) approaching the GBM CDF (dashed). Bottom right: $D_n \propto 1/\sqrt{n}$.

Part V — Module 4: Risk optimization

5.1 The portfolio and its limit

Hold a fraction f of wealth in the stock and $1-f$ in the bank, and **rebalance** back to the fraction f every Δt. ($f>1$ means borrowing at rate r to buy extra stock; $f<0$ would be short-selling.) Over one step the stock's log-return is $Y_k=(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}\,Z_k$ with $Z_k\sim N(0,1)$ i.i.d. (GBM, Module 3), and the bank returns $\rho=e^{r\Delta t}-1$ (Module 1). Wealth evolves as

$$\frac{W_{k+1}}{W_k}=1+x_k,\qquad x_k=f\,(e^{Y_k}-1)+(1-f)\rho.$$

Theorem 5.1 (continuous-rebalancing limit). As $\Delta t\to 0$,

$$\ln\frac{W_T}{W_0}\;\Rightarrow\; g(f)\,T+f\sigma\sqrt{T}\,Z,\qquad Z\sim N(0,1),\qquad \boxed{g(f)=r+f(\mu-r)-\tfrac{1}{2}f^2\sigma^2}.$$

Proof. All expansions are in powers of $\sqrt{\Delta t}$; $Y_k=O(\sqrt{\Delta t})$. By Lemma 1.6, $e^{Y_k}-1=Y_k+\frac{1}{2}Y_k^2+O(\Delta t^{3/2})=(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}Z_k+\frac{1}{2}\sigma^2Z_k^2\Delta t+O(\Delta t^{3/2})$, and $\rho=r\Delta t+O(\Delta t^2)$. Then $x_k^2=f^2\sigma^2Z_k^2\Delta t+O(\Delta t^{3/2})$ and $\ln(1+x_k)=x_k-\frac{1}{2}x_k^2+O(\Delta t^{3/2})$, so

$$\ln(1+x_k)=r\Delta t+f(\mu-r)\Delta t+f\sigma\sqrt{\Delta t}Z_k\underbrace{-\tfrac{1}{2}f\sigma^2\Delta t+\tfrac{1}{2}f\sigma^2Z_k^2\Delta t}_{\text{from the stock}}\underbrace{-\tfrac{1}{2}f^2\sigma^2Z_k^2\Delta t}_{\text{from }\ln}+O(\Delta t^{3/2}).$$

Sum over the $n=T/\Delta t$ steps: (i) $\sum Z_k^2\Delta t=T\cdot\frac{1}{n}\sum Z_k^2\to T$ by the law of large numbers; (ii) $\sum f\sigma\sqrt{\Delta t}Z_k\sim f\sigma\sqrt{T}\,Z$ exactly (a sum of independent normals); (iii) $n\cdot O(\Delta t^{3/2})=O(\sqrt{\Delta t})\to 0$. The two "stock" terms cancel in the limit, leaving $\ln(W_T/W_0)\to\big(r+f(\mu-r)-\frac{1}{2}f^2\sigma^2\big)T+f\sigma\sqrt{T}Z$. ∎

This is Itô's lemma in elementary form: the $Z_k^2\Delta t$ terms do not vanish (their sum tends to T), which is why a σ^2 correction appears. The $-\frac{1}{2}f^2\sigma^2$ term is the volatility drag of the whole portfolio.

5.2 Why $g(f)$ is "the growth rate"

From Theorem 5.1, $W_T=W_0\exp\big(g(f)T+f\sigma\sqrt{T}Z\big)$, so

$$\frac{1}{T}\ln\frac{W_T}{W_0}=g(f)+\frac{f\sigma Z}{\sqrt{T}}\;\xrightarrow[T\to\infty]{}\;g(f).$$

Over a long horizon wealth grows like $e^{g(f)T}$, whatever the luck. Also $g(f)T=E[\ln(W_T/W_0)]$, the expected log-wealth.

5.3 Maximum growth: the Kelly fraction

g is a quadratic polynomial, so it is continuous and differentiable:

$$g'(f)=\lim_{h\to 0}\frac{g(f+h)-g(f)}{h}=(\mu-r)-f\sigma^2,\qquad g''(f)=-\sigma^2<0.$$

g is strictly concave, so the unique maximum is where $g'(f)=0$:

$$f^* = \frac{\mu - r}{\sigma^2} = \frac{0.07}{0.04} = 1.75, \qquad g(f^*) = r + \frac{(\mu - r)^2}{2\sigma^2} = 9.125\%.$$

This is the **Kelly (Merton) fraction**. Note $g(f) - r = f\big((\mu - r) - \tfrac{1}{2} f\sigma^2\big)$ vanishes at $f = 0$ and $f = 2f^* = 3.5$: beyond $3.5\times$ leverage you grow *slower than the bank*, despite taking huge risk.

The derivative as a limit, numerically

Because g is quadratic, the difference quotient can be computed exactly: $\frac{g(f+h)-g(f)}{h} = g'(f) - \frac{\sigma^2}{2}h$. At $f = 1$ (where $g'(1) = 0.03$):

h	0.1	0.01	0.001	10^{-4}	10^{-5}	10^{-6}
quotient	0.028	0.0298	0.02998	0.029998	0.0299998	0.02999998

The error $-0.02h$ goes to 0 linearly in h. The program also finds f^* without the formula: g' is continuous with $g'(0) = 0.07 > 0$ and $g'(10) = -0.33 < 0$, so by the IVT bisection on g' finds the root 1.750000. (It uses the central quotient $\frac{g(f+h)-g(f-h)}{2h}$, which is *exact* for quadratics.)

The problem with Kelly: $f^* = 1.75$ means borrowing 75% of your wealth, and (see below) its one-year 95% VaR is 38.4%.

5.4 Measuring risk: Value-at-Risk

Definition 5.2. The one-year loss (as a fraction of initial wealth) is $\mathcal{L} = 1 - W_T/W_0$. The **Value-at-Risk** at level α is the loss that is exceeded only with probability $1 - \alpha$: $P(\mathcal{L} > \mathrm{VaR}_\alpha) = 1 - \alpha$.

Proposition 5.3. For $f \geq 0$,

$$\mathrm{VaR}_\alpha(f) = 1 - \exp\big(g(f)T - z_\alpha f\sigma\sqrt{T}\big), \qquad z_\alpha = \Phi^{-1}(\alpha).$$

Proof. VaR is the α-quantile of $\mathcal{L}$, i.e. $1-$(the $(1-\alpha)$-quantile of W_T/W_0). $W_T/W_0 = \exp(Y)$, $Y \sim N(g(f)T, f^2\sigma^2 T)$, and $\exp$ is increasing, so its $(1-\alpha)$-quantile is $\exp(g(f)T + z_{1-\alpha} f\sigma\sqrt{T})$ with $z_{1-\alpha} = -z_\alpha$. ∎

$z_{0.95} = 1.644854$ is found by bisection on $\Phi(z) - 0.95$ over $[-10, 10]$ — another IVT application. Note $\mathrm{VaR}(0) = 1 - e^{rT} = -3.05\%$: the all-bank portfolio has a negative VaR (a guaranteed gain).

5.5 The risk constraint: continuity and the IVT

The rule is $\mathrm{VaR}(f) \leq L = 0.15$. Define

$$h(f) = \mathrm{VaR}(f) - L.$$

- **Continuity:** h is a composition of a polynomial and $\exp$, so it is continuous on $[0, 2]$.
- **Signs:** $h(0) = (1 - e^{0.03}) - 0.15 = -0.1805 < 0$ and $h(2) = 0.4333 - 0.15 = 0.2833 > 0$.

By the Intermediate Value Theorem there exists $f_{VaR} \in (0, 2)$ with $h(f_{VaR}) = 0$: an allocation whose risk is *exactly* at the limit.

Proposition 5.4 (uniqueness and the feasible set). If $(\mu - r)\sqrt{T} < z_\alpha \sigma$, then $\mathbf{VaR}$ is strictly increasing on $[0, \infty)$. Hence the root f_{VaR} is unique and $\{f \geq 0 : \mathbf{VaR}(f) \leq L\} = [0, f_{VaR}]$.

Proof. $\mathbf{VaR}(f) = 1 - e^{\eta(f)}$ with $\eta(f) = rT + f\big((\mu - r)T - z_\alpha \sigma \sqrt{T}\big) - \frac{1}{2} f^2 \sigma^2 T$.
$\eta'(f) = (\mu - r)T - z_\alpha \sigma \sqrt{T} - f\sigma^2 T < 0$ for $f \geq 0$ under the assumption, so η decreases and $\mathbf{VaR}$ increases. Here $(\mu - r)\sqrt{T} = 0.07 < z\sigma = 0.329$. ∎

Bisection on h over $[0, 2]$

k	a_k	b_k	midpoint	h(mid)	keep
1	0	2	1	+0.0704	left half
2	0	1	0.5	−0.0508	right half
3	0.5	1	0.75	+0.0109	left half
4	0.5	0.75	0.625	−0.0196	right half
5	0.625	0.75	0.6875	−0.0043	right half
6	0.6875	0.75	0.71875	+0.0034	left half
7	0.6875	0.71875	0.703125	−0.0005	right half
8	0.703125	0.71875	0.710938	+0.0015	left half
⋮					
41	0.705014	0.705014	**0.705014**	10^{-13}	stop

Cross-check with a closed form

In this model $h(f) = 0$ is a quadratic equation: $\eta(f) = \ln(1 - L)$, i.e. $\frac{1}{2}\sigma^2 T f^2 + \big(z_\alpha \sigma \sqrt{T} - (\mu - r)T\big) f + \big(\ln(1 - L) - rT\big) = 0$, i.e. $0.02 f^2 + 0.258971 f - 0.192519 = 0$, whose positive root is $f = 0.7050142$ — identical to bisection. **Why use bisection then?** Because it only needs continuity. With realistic risk models (historical returns, fat-tailed distributions, Monte Carlo VaR) there is no formula, but the IVT argument and bisection still work unchanged.

5.6 The optimal allocation

Problem: maximise $g(f)$ subject to $0 \leq f \leq f_{max}$ and $\mathbf{VaR}(f) \leq L$. By Proposition 5.4 the feasible set is the closed interval $[0, c]$ with $c = \min(f_{VaR}, f_{max})$. Since g is continuous, the Extreme Value Theorem guarantees a maximiser exists. Since g is concave with peak at f^*, the maximiser on $[0, c]$ is f^* if $f^* \leq c$, and otherwise c (because $g' > 0$ on $[0, f^*)$, g is increasing there):

$$f_{opt} = \min\left(f^*,\, f_{VaR},\, f_{max}\right) = \min(1.75,\ 0.705,\ 2) = 0.705.$$

Invest 70.5% in the stock and 29.5% in the bank. Growth $g(0.705) = 6.94\%$ per year (vs. 9.13% for Kelly) with VaR exactly 15% (vs. 38.4% for Kelly). The risk constraint is *binding*.

Sensitivity: how the answer depends on the inputs

loss limit L	5%	10%	15%	20%	25%	40%
f_{opt}	0.307	0.503	**0.705**	0.913	1.128	1.750
growth $g(f_{opt})$	4.96%	6.02%	6.94%	7.72%	8.35%	9.13%

More risk tolerance allows more stock and more growth. At $L = 40\%$, $f_{VaR} = 1.83 > f^*$: the constraint is no longer binding and the answer becomes Kelly — the "min" switches branch. With $L = 15\%$, higher volatility lowers the allocation: $\sigma = 15\% \Rightarrow f_{opt} = 1.023$; $\sigma = 25\% \Rightarrow 0.538$; $\sigma = 30\% \Rightarrow 0.435$.

5.7 Verification by Monte Carlo simulation

The program simulates 20,000 one-year wealth paths with discrete rebalancing (the model of §5.1, before the limit) and computes the empirical 5% quantile of the loss. By the law of large numbers (more precisely Glivenko–Cantelli), the empirical quantile converges to the true one, with standard error

$$\mathrm{SE} \approx \frac{\sqrt{\alpha(1-\alpha)/N}}{\text{density of } W_T/W_0 \text{ at the quantile}} = \frac{\sqrt{0.05 \cdot 0.95/20000}}{0.861} \approx 0.18\%.$$

rebalancing	yearly	quarterly	monthly	weekly	daily
simulated VaR	14.63%	15.06%	14.76%	14.89%	14.91%

All values are within about two standard errors of the analytical 15%; yearly rebalancing additionally differs from the continuous model because $\Delta t = 1$ is far from the limit. The simulation confirms Theorem 5.1 and Proposition 5.3.

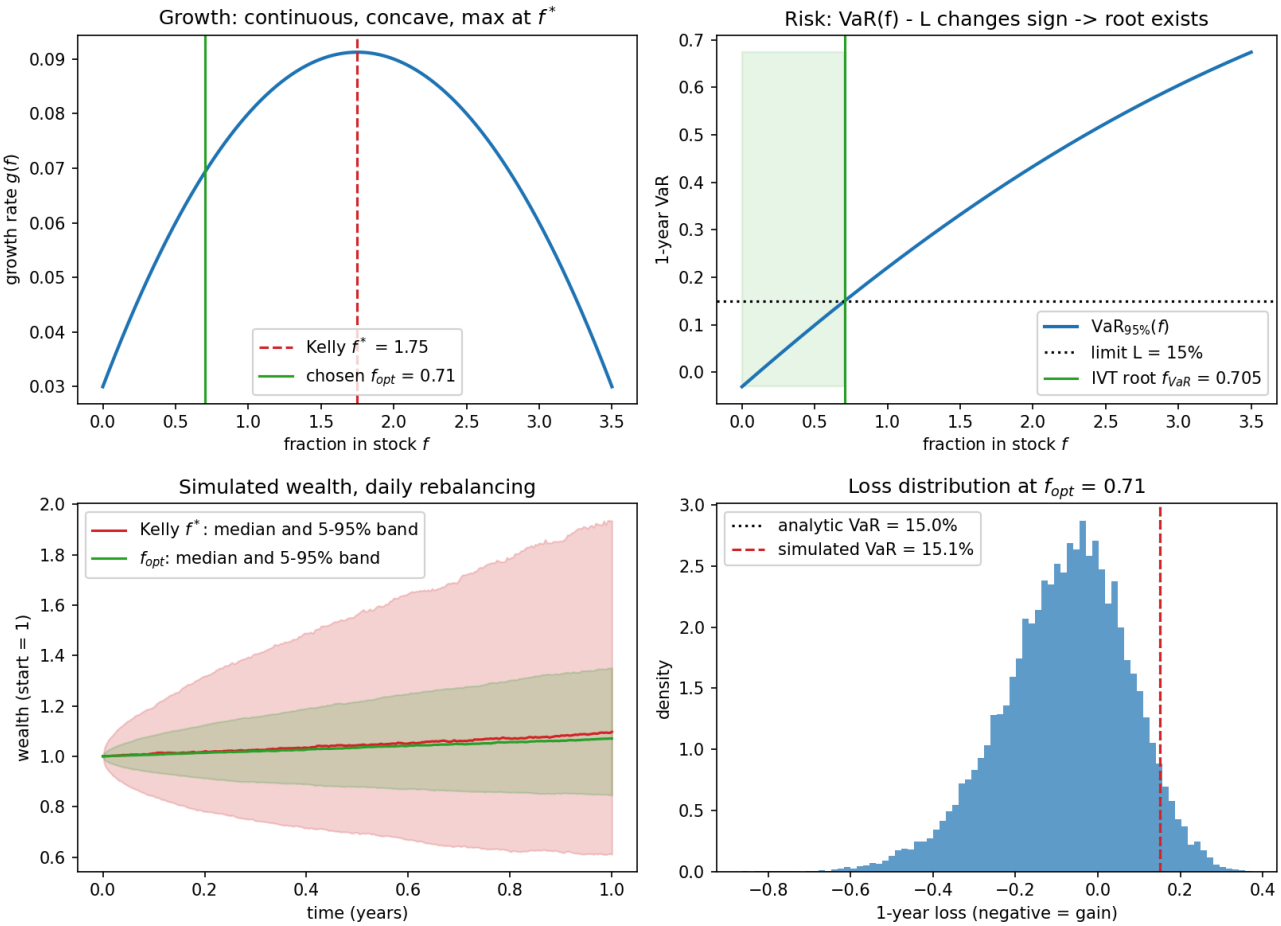


Figure 4. Top left: concave growth $g(f)$, max at Kelly f^*. Top right: VaR(f) crosses the limit L at the IVT root. Bottom left: simulated wealth bands — Kelly is much wider. Bottom right: simulated loss distribution at f_{opt}.

Part VI — From mathematics to code

Mathematics	Python (`src/`)	MATLAB (`matlab/`)
Bisection (Thm 1.10)	`bisection.bisection`	`bisection.m`
Φ, φ, Φ^{-1}	`normal.norm_cdf / norm_pdf / norm_ppf`	`norm_cdf.m` …
A_n, Pe^{rt}, error (3)	`compounding.compound / continuous / error_estimate`	`compound.m` …
CRR tree, Black–Scholes	`option_pricing.binomial_price / black_scholes`	`binomial_price.m` …
Walk distribution, D_n	`gbm.walk_log_return_distribution / ks_distance_to_gbm`	`ks_distance_to_gbm.m` …
g, f^*, VaR, f_{VaR}, f_{opt}	`risk.growth_rate / kelly_fraction / value_at_risk / max_fraction_under_var / optimal_fraction`	same names `.m`
Monte Carlo §5.7	`risk.simulate_wealth / empirical_var`	`simulate_wealth.m`, `quantile_linear.m`

Numerical details worth knowing

- **$\log 1p$ for compounding.** For $n = 10^7$, $1 + r/n = 1.00000001$ loses digits in floating point. The code computes $\exp(nt \cdot \log 1p(r/n))$, where `log1p(u)` evaluates $\ln(1+u)$ accurately for tiny u.
- **Logarithms for binomial probabilities** (§4.3) avoid overflow of $\binom{n}{k}$.
- **No statistics libraries.** $\Phi(x) = \frac{1}{2}\left(1 + \operatorname{erf}(x/\sqrt{2})\right)$ and Φ^{-1} by bisection, so the code is self-contained and identical in Python and MATLAB.
- **Exactness vs. simulation.** Modules 1–3 and the optimisation are deterministic (Python and MATLAB agree to many digits). Only the Monte Carlo checks are random; they differ slightly between languages.

Part VII — Summary, assumptions, and likely questions

7.1 Where each theorem is used

Mathematical idea	Where
Limit of a sequence, squeeze, monotone convergence	$(1+r/n)^{nt} \to e^{rt}$ (Thm 2.1, Prop 2.2)
Continuity: limit passes inside **exp**	Thm 2.1, Thm 5.1, every exponential model
Taylor expansion / asymptotics	rate $1/n$ (§2.3), expansion of p (§3.4), Thm 5.1
Limit in distribution, CLT	binomial tree → Black–Scholes; random walk → GBM
Rates of convergence	$1/n$ (compounding), $1/N$ (options), $1/\sqrt{n}$ (KS distance)
Derivative as a limit; concavity	$g'(f)$, Kelly fraction f^*
Intermediate Value Theorem	existence of f_{VaR}, of $z_{0.95}$, and of the root of g'
Bisection (nested intervals)	computing all three roots; 41 iterations
Extreme Value Theorem	existence of the optimal allocation on $[0, c]$
Law of large numbers	$\sum Z_k^2 \Delta t \to T$ (Thm 5.1); Monte Carlo check

7.2 Assumptions and limitations

- Constant μ, σ, r; in reality they change and μ is hard to estimate.
- Normal log-returns. Real returns have *fat tails*, so the true 95% VaR is somewhat higher than the model's.
- Free, continuous rebalancing; borrowing at the risk-free rate; no taxes.
- VaR ignores *how bad* losses beyond the 5% threshold are (Expected Shortfall addresses this). The same IVT/bisection method works for it, since it is also continuous in f.

7.3 Questions you may be asked

Where exactly is the limit in your project?

Four places: $(1+r/n)^{nt} \to e^{rt}$; the binomial price $C_N \to C_{BS}$; the random walk → GBM as $\Delta t \to 0$; and the derivative $g'(f)$ as the limit of difference quotients. Theorem 5.1 (discrete rebalancing → continuous) combines them.

Where is continuity used, and why is it essential?

The VaR function is continuous in f, with $h(0) < 0 < h(2)$, so the IVT guarantees an allocation with risk exactly 15%. Without continuity (for example, if risk could jump) the root might not exist and bisection could fail.

Why not just solve the quadratic?

Here we can, and it confirms 0.705014. But bisection only needs continuity, so it still works for realistic risk measures that have no formula.

How do you know bisection stops after 41 steps?

The interval halves each time: after k steps its length is $2/2^k$. We need $2/2^k < 10^{-12}$, i.e. $k > \log_2(2 \cdot 10^{12}) \approx 40.9$.

Why not invest the Kelly fraction 1.75?

It needs 75% borrowed money and has a 38.4% one-year VaR. Growth is a flat hill near the top: going from 1.75 to 0.705 reduces growth only from 9.13% to 6.94%, but cuts the risk from 38.4% to 15%.

What is the $-\frac{1}{2}\sigma^2$ term?

Volatility drag: the average price grows at μ, but the typical (median, log) growth is $\mu - \frac{1}{2}\sigma^2$, because losses hurt more than equal gains help (-50% needs $+100\%$ to recover). It is also why growth falls when you take too much risk.

Why does the binomial price oscillate?

The strike's position relative to the tree's final nodes alternates with the parity of N, so the error constant alternates ($N \cdot$error ≈ 2.00 for even N, 1.75 for odd N). Both still go to zero like $1/N$.

How did you check the results are right?

15 automated tests compare against known values ($e^{0.1}$, the Black–Scholes reference 10.4506, put–call parity, $z_{0.95}$, $f^* = 1.75$), the closed-form root confirms bisection, the $1/\sqrt{n}$ prediction matches the KS distances, and Monte Carlo simulation confirms the VaR.

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